SAVESPOTS RESEARCH DEPARTMENT | 2026 COHORT

Deliverables, Standards, and Semester Timeline

Revision 2 — what you build, how you build it, and when it is due

Reference paper: *Ilyas, A. I., & Ilyas, S. (2026). Structural Gaps in Urban Harm Reduction Infrastructure: A Block-Level Access and Policy Analysis of Dallas County, Texas*

Reference repository: github.com/Ilyas1434/dallas-research-savespots

Archive: doi.org/10.5281/zenodo.21537399

Program window: Kickoff Tuesday, July 28, 2026. Fifteen Sunday meetings, August 2 through November 8, 2026.

> **THIS DOCUMENT SUPERSEDES OUR OLD DELIVERABLES GUIDE**

00 · HOW TO USE THIS DOCUMENT

Read Sections 1 through 3 before Sunday. They define the output chain, the standards, and how you choose your analysis. Sections 4 through 11 are reference material for you to read; reach each the week before that deliverable is due. Section 12 is the calendar; keep it open.

The Dallas paper is your benchmark, not your template. It is a demonstration of what depth, honesty about limitations, and analytical integration look like at this program's standard. Your city is NOT Dallas. Your paper should do for your city what that paper does for Dallas — which in most cases means a different analysis, not the same one with different ZIP codes.

01 · WHAT YOU PRODUCE

#	Deliverable	What it becomes	Due
1	City Profile & Harm Reduction Landscape	Manuscript §3 and §5	Week 3
2	Policy Matrix	Manuscript §4	Week 4
3	Data Inventory & Analysis Plan	Manuscript §2; scoping gate approval	Week 4
4	Analytic Repository (GitHub + Zenodo DOI)	Reproducibility backing for every number you publish	Week 10
5	Manuscript	The paper	Week 13
6	Preprint (medRxiv)	A citable DOI with your name on it	Week 15
7	Conference Abstract	Symposium submissions	Week 11 draft, Week 15 final
8	Poster	What you present	Week 15

Optional, distinction-level: a stakeholder leave-behind brief and a deployed interactive map. Neither is required to complete the program.

02 · THE RESEARCH CONTRACT

2.1 Classify every quantity

Every number in your paper is one of three things, and you state which wherever ambiguity could arise:

Class	Definition	Example from the reference paper
Measured	A direct count of people or a computed geometric quantity. No modeling, interpolation, or smoothing.	2,613,539 residents (2020 decennial count); distance from each census block to the nearest supply point
Modeled	A published model-based rate, a survey estimate, or any index derived from them.	CDC NCHS model-based tract overdose rates; ACS 5-year poverty and uninsurance estimates; the composite vulnerability score
Published	A figure transcribed from a cited report, with the page or figure reference.	558 calendar-year 2024 deaths, from the DCHHS OD2A annual surveillance report

This is the what makes a student paper read as careful rather than credulous.

2.2 Name your primary measured quantity

At least one of your paper's three core answers must rest on something you measured, not something you modeled. State in your methods what that quantity is and why it is the defensible target given your city's data environment.

In Dallas that quantity is block-level distance to naloxone supply, chosen because mortality was suppressed and distance was not. In a city with open point-level death records (for example, Chicago), your measured quantity might be observed overdose density itself. In a city with neither, it might be the supply registry you build. The choice differs; the requirement does not.

2.3 Disclose weights; run sensitivity

If you build any composite, index, or ranking, you must disclose the weighting scheme, justify it, and re-run the pipeline under at least two alternative weightings. Report the share of units that change classification.

Report the result even when it fails!! The reference paper's distance-weighted variant relocated 21.5% of tracts, above its own 15% stability threshold, and says so in the abstract-adjacent methods and again in the limitations: "the Tier 1 set is one defensible configuration, not a robust one." Disclosure ALWAYS strengthens research.

2.4 Report gaps as gaps

Where a quantity cannot be produced from the data you have, report it as not computable and attribute it to the source that would hold it.

Two examples from the reference paper: concordance between decedent residence and overdose incident location, and the proportion of deaths occurring within walking distance of supply. Both require case-level medical examiner records and are recorded as gaps.

2.5 Never allocate a coarse measure to a fine geography

If a statistic is published only at county level, it stays at county level. Pushing a county figure down to census tracts to fill a hole in your map is fabrication.

The reference paper excluded two layers it wanted — non-fatal overdose emergency department visits and point-in-time homelessness counts — for exactly this reason, and documented the exclusion as a limitation rather than working around it.

2.6 Take the conservative figure

Every key statistic — mortality figures, policy legality claims, program reach — is traceable to a primary source, or to a secondary source that cites a primary source. Where two credible sources conflict, present both, explain the likely reason for the discrepancy, and use the more conservative estimate. Say that you did. This is the correct scientific move, not a hedge.

2.7 Reproducibility

Every headline number in your paper is reproduced by running one command in your repository. If a reader cannot regenerate a figure from your code and cited inputs, that figure does not go in the paper.

2.8 The privacy floor

> **HARD RULE — NO EXCEPTIONS**
>
> Nothing you publish — no figure, no table, no map, no repository file, no poster panel — may contain individual-level records, addresses, or coordinates of overdose events or decedents.
>
> This matters most in cities with open point-level medical examiner data, where the temptation to plot the points is strongest. You may analyze at point level locally. You publish aggregated only: tract, ZIP, community area, etc.
>
> Preprint servers screen for participant-identifying information, and a violation is a decline. More to the point, it is a harm to the communities this program exists to serve.

03 · CHOOSING YOUR ANALYSIS

Your city's data environment determines what analysis makes sense.

3.1 Three questions every paper answers

The methods are yours. These three questions are not. Your paper answers all three, in this order, using whatever approach your data supports.

Question	What it means	How Dallas answered it
1. Where is the burden?	Describe overdose burden at the finest resolution your data permits — and state what that resolution is.	Section 8.7: mortality at 150 CDC aggregation units, 92% with suppressed counts, presented explicitly as the evidentiary limit rather than as a prioritization tool.
2. Where is the gap?	Identify where need and available service diverge, using a quantity you can measure.	Sections 8.1–8.5: measured block-level access deficit plus a five-layer composite with disclosed weights and reported sensitivity.
3. Where should supply go?	Produce an actionable, ranked, geographically specific output. Not a list of neighborhoods, but named places with a stated selection rule.	Section 8.6: 5,428 candidate sites ranked by need × measured reach, with a de-overlapped greedy coverage analysis reporting marginal gain at 10, 25, and 50 sites.

3.2 Four data environments

Declare which of these describes your city in Deliverable 3. Most cities will be a hybrid of two.

Environment A — Suppressed mortality

Fine-geography death counts are withheld under state confidentiality rules; no usable sub-county mortality product exists. This is Dallas, and it is the most common environment nationally.

- Your measured quantity is access, coverage, or distance — something geometric that suppression cannot touch.
- Your contribution is method under constraint: you demonstrate that a defensible priority ranking can be produced without the signal everyone assumes is required.
- The strongest available proof is an independence check — build your ranking twice, once with the modeled mortality layer and once without it, and report the rank correlation. Dallas found $\rho = 0.966$, which is why the recommendation survives the data environment rather than depending on it.

Environment B — Open point-level mortality

The county medical examiner publishes case-level overdose records with geography. Cook County, Illinois is the clearest example; several other large counties publish comparable archives.

- You can run the validation Dallas could not: does an access-based or vulnerability-based prioritization actually align with observed death geography? Answer that and you have a genuinely novel result, not a replication.
- This is the richest environment and carries the strictest privacy obligation. See Section 2.8. Aggregate before you publish anything.

- Beware the residence-versus-incident distinction. Records may geocode to place of death, place of injury, or decedent residence, and these are different questions. State which one you used and what it can and cannot support.

Environment C — No supply directory

No state or county publishes a machine-readable inventory of naloxone access points. You will assemble one from scattered pharmacy lists, health department pages, nonprofit sites, and phone verification.

- Treat this as a data contribution. The registry you build is a publishable artifact in its own right, and it is the input every other analysis in your paper depends on.
- Document the compilation protocol in your methods with the same rigor you would document a statistical procedure: what sources you searched, what your inclusion criteria were, how you geocoded, how you handled duplicates, what your verification rate was, and what you could not verify.
- Record each site's access reality, not just its existence — walk-in versus appointment, hours, whether it dispenses to the public or only to enrolled patients. Dallas's headline access figure is deliberately conservative because it credits as supply 13 opioid treatment programs that dispense methadone to enrolled patients and 7 nightlife venues open only during business hours. Say the same thing about yours.

Environment D — Rich non-fatal data

Your city publishes EMS naloxone administrations, ODMAP participation, or emergency department overdose visits at sub-county geography.

- You can include the layer Dallas had to exclude. Non-fatal events typically occur at several times the fatal count, so this is a materially better-powered signal than mortality in most cities.
- Do not treat non-fatal data as a clean substitute for mortality. It reflects where an ambulance was called, which is itself shaped by transport access, Good Samaritan protection, and community trust in police response. That confounding is analytically interesting and belongs in your paper.

> **WHAT IS NOT ACCEPTABLE**
>
> Choosing an environment because it is easier. Your declaration in Deliverable 3 must be evidenced — you show what you searched for and what you found or did not find.
>
> For example, Reproducing the Dallas analysis in a city where better data exists. If your medical examiner publishes point-level records and you build an access-only proxy anyway, you have chosen the weaker instrument and the paper will say so.
>
> Another example: a generic composite index with undisclosed weights and no sensitivity analysis. This is the single most common failure mode in student spatial work and it is disqualifying under Section 2.3.

04 · DELIVERABLE 1 — CITY PROFILE & HARM REDUCTION LANDSCAPE

Due Week 3 (Aug 16). Becomes manuscript §3 and §5. Target 1,500–2,500 words.

This is your first substantive deliverable and it is deliberately pipeline-independent — you write it while your data inventory is still in progress. It exists to establish whether your city has a story worth telling before anyone builds infrastructure.

It is not a census summary. Every statistic in it connects to overdose risk, naloxone access, treatment access, or a policy barrier.

4.1 Required content

Element	What it must contain	Reference standard
Overdose burden	Mortality trend across at least three years with percent change, fentanyl contribution, polysubstance share, and non-fatal volume if published. Attribute each figure to its source and window.	Dallas: 77.1% increase in overdose deaths 2016–2024; fentanyl-related deaths up 1,550% over the same period.
Demographics with implications	Population, racial and ethnic composition, poverty, uninsurance — each with a stated harm reduction consequence, not reported bare.	Dallas: 21% uninsured, among the highest of large U.S. urban counties, which directly constrains medication-assisted treatment access given Medicaid non-expansion.
Treatment and infectious disease	Treatment capacity and retention; HIV and hepatitis C burden as an injection-drug-use proxy.	Dallas §3.4–3.5.
Criminal justice intersection	How possession enforcement and incarceration interact with treatment-seeking and emergency contact behavior.	Dallas §3.6.
Existing infrastructure	Named organizations, what each actually does, and which geographies each reaches. Not a directory, an assessment.	Dallas §3.7: distribution model, supply backbone, and peer outreach identified by organization and function.
Who is missed	The populations current outreach systematically fails to reach, and why. This is the part that separates strong work from acceptable work.	Dallas §5.5: people who are young, employed, socially integrated, and do not self-identify as drug users are not reached by outreach targeting encampments.
The single most critical gap	One sentence naming the most consequential structural gap in your city's harm reduction system.	Dallas: no embedded, non-institutional naloxone model capable of reaching people who do not self-identify as drug users.

4.2 The analytical move

Every paragraph does four things in sequence: states a local condition, gives the supporting evidence, interprets what it means, then draws the implication for harm reduction access or delivery in your city. A paragraph that summarizes a source without interpreting it is a weak paragraph no matter how well written.

> **THE SUBSTITUTION TEST**
>
> **Fails:** *"Fentanyl is increasingly contaminating the drug supply across many American cities, creating overdose risks for populations who may not know they are consuming opioids."*

> **Passes:** *"Opioid-stimulant combination deaths rose 260% in Dallas County between 2018 and 2024, reflecting a supply in which fentanyl is now ubiquitous across substance categories. The people at risk in these contexts are largely young, employed, and socially integrated; they do not present at harm reduction services and are not reached by outreach targeting homeless encampments."*

05 · DELIVERABLE 2 — POLICY MATRIX

Due Week 4 (Aug 23). Becomes manuscript §4.

A structured analytical comparison of every major policy shaping harm reduction in your jurisdiction. It is not a list of laws. For every policy you answer two questions: what does the law say, and what does it actually do? The distance between those answers is where the analytical content lives.

5.1 Required columns

Column	What goes here
Policy	Name of the law, ordinance, or standing order
Level	City / County / State / Federal
Status (as of date)	Active / Partial / Pending / Defeated / Not enacted — always with the date of assessment
Real-world impact	What it does on the ground: enforcement variability, access gaps, and who it reaches or systematically excludes
Strategic implication	What it permits, what it blocks, and what opportunity it creates for community naloxone distribution

5.2 Minimum coverage

- Naloxone access: standing orders, over-the-counter status, pharmacy dispensing rules, and third-party prescription authority
- Good Samaritan law: scope of protection, exclusions, enforcement reliability, and actual effect on bystander 911 calls
- Fentanyl test strip legality: current classification and the status of any pending legislation
- Syringe services programs: authorization status and any county-level carve-out or pilot provision
- Medication-assisted treatment environment: methadone clinic requirements and buprenorphine prescribing capacity after elimination of the DATA 2000 waiver
- Drug possession penalties: penalty structure and its effect on treatment-seeking and emergency contact
- Medicaid expansion status and what it specifically means for opioid use disorder treatment access in your county
- Opioid settlement funds: allocation amounts and spending breakdown, particularly the share directed to harm reduction versus treatment versus uncommitted

5.3 The two-layer standard, illustrated

"Good Samaritan Law: Active" is not an entry. The reference paper's treatment of the Jessica Sosa Act shows the standard: it identifies the statute correctly (Texas Health & Safety Code § 481.115(g)–(i)), establishes that it provides a restricted affirmative defense rather than genuine immunity, explains which categories of caller

are excluded from that defense, and connects the resulting deterrence to preventable mortality. Then it names the specific legislative fix — replacement modeled on Pennsylvania's Act 139 — drawn from the comparative case in §6.

Get statutory citations exactly right. Cite the code section, not a news article's paraphrase of it. A wrong statutory citation in a published preprint is permanent.

06 · DELIVERABLE 3 — DATA INVENTORY & ANALYSIS PLAN

Due Week 4 (Aug 23). This is the scoping gate. You do not build until it is approved.

The most expensive mistake available to you this semester is building a pipeline for an analysis your city's data cannot support. This deliverable exists to catch that in week 4 rather than week 9.

6.1 The inventory table

One row per dataset you investigated — including the ones you could not get. A documented absence is a finding; a silent omission is not.

Column	What goes here
Dataset	Name and publisher
Endpoint	URL, API identifier, or "PDF only" / "request required" / "does not exist"
Resolution	The finest geography at which it is actually published — not the geography you wish it had
Vintage	Most recent record date and the date you retrieved it
Access	Open / keyed / request / unavailable
Verdict	Using it, excluding it (with reason), or reporting it as a gap

6.2 Minimum layers to investigate

These exist nationally and you are expected to have checked all of them. The first five are available for every U.S. county.

- Census 2020 P.L. 94-171 block population (variable P1_001N) and TIGER/Line block and tract geography
- American Community Survey 5-year subject tables — poverty (S1701) and uninsurance (S2701) at minimum
- CDC/ATSDR Social Vulnerability Index
- CDC NCHS drug overdose mortality products, at whatever geography they publish for your county
- SAMHSA treatment facility locator
- Your transit agency's GTFS feed, if one is published
- Your state or county naloxone access point registry, if one exists — and if not, say so explicitly and move to Environment C
- **Your county medical examiner or coroner's office, for case-level or aggregated overdose records – this is literal gold!**
- Your city or county open data portal, for business licenses, points of interest, libraries, and code or blight complaints
- Your state and county health department dashboards and annual surveillance reports

6.3 The plan itself

Two pages maximum, answering six questions:

1. Which data environment (A–D, or which hybrid) describes your city, and what evidence supports that declaration?

2. What is your primary measured quantity, and why is it the defensible target here?
3. How will you answer each of the three questions in Section 3.1?
4. What is your geographic unit, and why is it the finest one your data honestly supports?
5. If you are building a composite: what layers, what weights, what justification, and what sensitivity variants will you run?
6. What will you not be able to compute, and how will you report it?

> **GATE CONDITION**
> Ali and Sameer will sign off on this plan before you write pipeline code. An approved plan can still change — analysis plans always do — but a change after Week 6 requires a note in your methods explaining what changed and why. That note is normal scientific practice, not a penalty.

07 · DELIVERABLE 4 — THE ANALYTIC REPOSITORY

Built Weeks 5–9. Frozen Week 10 (Oct 4). One repository and one Zenodo DOI per city.

Your repository is not a folder of scripts. It is the evidence that your numbers are real. Study the reference repository before you build yours — it is the structural benchmark.

7.1 Required structure

Path	Contents
scripts/	One script per pipeline stage, plus a single run_all script that executes them in dependency order
data/raw/	Dated snapshots of every input, so a published run can be reproduced after upstream data changes
data/clean/	Every derived output, including the metadata files that record row counts, date ranges, and null rates
figures/	Every manuscript figure at 300 dpi in both PNG and PDF, exported by the pipeline rather than by hand
README.md	What the analysis does, why it is built that way, headline outputs with the file each comes from, every data source with vintage and access status, and the full limitations list
report.md or insights.md	Dense, quantitative findings with every claim sourced. This is what you write the manuscript from.
LICENSE / LICENSE-DATA	MIT for code; CC BY 4.0 for derived data outputs; upstream inputs retain their publishers' terms
CITATION.cff	So the repository is citable and Zenodo metadata is clean

7.2 Standards

- Every script logs row counts, date ranges, and null or geocode-failure rates. If your geocode failure rate exceeds 10%, stop and raise it at the next Sunday meeting.
- Handle suppression explicitly. Record suppressed units and carry them into your limitations. Never silently impute.

- No secrets in the repository. API keys live in a gitignored .env with a committed .env.example.
- Pin your run date so an archived run can be reproduced exactly.
- Scan your commit history before making the repository public. Making a repository public exposes every past commit, not only the current tree — if a key was ever committed and later removed, it is still in the history.
- Do not commit data you are not licensed to redistribute. Cached commercial API responses and republished copyrighted PDF reports are the two common traps. Cite the source and link it instead; the figure you quote from a report is a fact, and facts are not copyrightable, but the document is.

7.3 Freeze and archive

At Week 10 you tag a release, deposit it to Zenodo, and mint a DOI. From that moment your numbers do not move. Every figure in the manuscript, every statistic in the abstract, and every panel on the poster comes from the frozen run.

08 · DELIVERABLE 5 — THE MANUSCRIPT

Full draft due Week 13 (Oct 25). APA 7th edition.

8.1 Format

- APA 7th edition throughout: in-text citations, reference list, headings, title page, abstract
- Times New Roman 12pt, double-spaced, one-inch margins
- Minimum 20 credible sources; 25–35 is typical for a complete analysis
- Every figure embedded with an APA-compliant caption and a note listing every data source, vintage, and projection
- Length is whatever the analysis requires. The reference paper runs well past any page target because the work needed the space. Do not pad, and do not cut substantive analysis to hit a number.

8.2 Section structure

Section	What it must accomplish
1. Introduction	Opens with city-specific burden data in the first paragraph — never a national background that could appear in any paper. States objectives. Names the gap in the literature your paper addresses, including the methodological gap if you have one.
2. Methodology	Study design and the measured/modeled/published classification. Every data source with vintage and accession identifier. Your primary measured quantity and its justification. Composite construction with weights and sensitivity. A limitations subsection that is specific, not ritual.
3. City profile	From Deliverable 1. Demographics tied to harm reduction implications, overdose trend with year-by-year breakdown, infectious disease burden, criminal justice intersection, existing infrastructure by organization and reach.
4. Policy analysis	From Deliverable 2. Two layers for every policy: what the law says and what it does.
5. Implementation analysis	For each major program: who funds it, who delivers it, which communities it reaches, and which it systematically misses. Includes the cultural context of drug use in your city and the populations invisible to standard outreach.
6. Comparative case study	One structurally analogous case and one best-practice case, each with explicit selection justification. Extract named, transferable lessons — not a description of what other cities did. State plainly that the comparison is structured qualitative evidence, not causal inference.
7. Gap identification	Minimum six categories with sub-headers: policy, implementation, geographic, demographic, data, access, cultural, structural. Each gap specific and evidenced from the preceding sections.
8. Geographic prioritization	Your primary analysis. Tiered or ranked output, each unit named, each classification explained by the indicators that drove it, each paired with a recommended approach. Zones of structurally undetectable risk identified separately. Close with your burden data at its honest resolution.
9. Metrics and evaluation	Short- and long-term metrics, each tied to a gap in §7. For each: what it measures, why it matters in this city, its current baseline, and how it would be recomputed. State plainly what your metrics cannot attribute.
10. Recommendations	Each recommendation explicitly tied to a gap finding and, where geographic, to a mapped location. Each with a feasibility note given your city's actual policy constraints. Nothing appears that the analysis did not earn.

Section	What it must accomplish
11. Conclusion	Synthesizes rather than summarizes. Reinforces the central analytical argument. Introduces nothing new. Acknowledges the limitations that most bound the findings.
12. References	APA 7, alphabetical, hanging indent. Every in-text citation in the list and vice versa. No source you did not actually read.

8.3 Front matter you will need

Your title page carries: title, authors with superscript affiliations, the program identification line, and the date. Your author note carries the six statements below. Draft them at Week 14. THIS IS EXTREMELY IMPORTANT FOR PRE-PRINTING!

Statement	What it must say
Author contributions	Who did what, specifically. Analysis, drafting, revision, and final approval are named separately.
Ethics	That the study used only publicly available, de-identified, aggregate secondary data; that it does not constitute human subjects research; that no IRB approval was required; and that no individual-level records were obtained or analyzed at any stage.
Funding	Whether any external funding was received.
Competing interests	Your SaveSpots affiliation is disclosed in full. Say plainly that the analysis reports findings without recommending action by any specific organization — and then make sure the paper actually reads that way.
Data availability	Your repository URL and your Zenodo DOI, plus a statement that all primary inputs are public datasets identified in the methods.
Use of AI	That AI tooling was used for pipeline construction, figure generation, and drafting support; that all statistics derive from cited primary sources; and that you verified every reported value and take full responsibility for the content.

8.4 Neutrality

Preprint servers accept original research and decline commentary and opinion. A paper written as "SaveSpots should do X" can read as advocacy and be declined; the same analysis written as findings reads as research. Address recommendations to public health authorities, community-based organizations, and legislative bodies generally, not to your own organization. Run one dedicated neutrality pass over your recommendations section before submission.

09 · DELIVERABLE 6 — THE PREPRINT

Submitted Week 15 (Nov 8). Screening typically takes 2–4 business days.

9.1 Before you submit

- Verify current requirements on the server's own pages the week you submit. Policies change; do not rely on this document or on secondhand summaries.
- Confirm your article type is accepted. Original research, systematic reviews, and methodological research are accepted; narrative reviews, commentary, and opinion are not.

- Confirm the manuscript has not been posted or published elsewhere. This sets a hard sequencing rule below.
- All six statements from §8.3 are in the manuscript, not just in the submission form.
- No participant-identifying content anywhere, including figures.
- Every co-author has approved the final version in writing.
- Every author has an ORCID ID. This takes two minutes and is free.
- Corresponding author uses given SaveSpots address (will be created iA)

> **SEQUENCING RULE — NON-NEGOTIABLE**
>
> The preprint posts before the paper, its PDF, or its figures appear anywhere else public — website, social media, anywhere. Preprint servers decline manuscripts already posted elsewhere. Once the DOI is live, link to it freely.
>
> Your interactive map, if you build one, is a separate artifact and may be deployed at any time. It links from the SaveSpots site alongside the DOI. It does not go in the preprint.

9.2 After it posts

A preprint is not prior publication at the large majority of journals, and your manuscript may change substantially — including its methodology — between the preprint and a journal submission. A revision posts under the same DOI at any point before journal acceptance, so improvements you identify later cost you nothing.

10 · DELIVERABLE 7 — THE CONFERENCE ABSTRACT

Drafted Week 11 (Oct 11) from frozen numbers. Final Week 15.

You will write two related but distinct abstracts. The manuscript abstract summarizes the paper for a research reader. The conference abstract is a submission — it competes for a slot against other submissions and is read by a reviewer deciding whether your work merits presentation. Write the manuscript abstract last; write the conference abstract from it.

10.1 Structure

Most symposia accept 250–300 words and either require or accept structured headings. Default to structured. Four moves, in order:

Move	Target	What it must contain
Background	3–4 sentences	Your city's burden in actual numbers, the policy environment in concrete terms, and the specific constraint or gap that motivates the analysis. Never open with national context.
Methods	3–4 sentences	Geographic unit, primary measured quantity, what was computed and how, weighting and sensitivity if you built a composite, and comparison cases if any. Enough that a reader knows whether to believe you.
Results	4–5 sentences	Your strongest measured findings with exact figures. Lead with the single most arresting number. This is the section reviewers actually read.
Conclusions	2–3 sentences	What follows, what is actionable now without policy change, and the one methodological point worth carrying forward.

10.2 Rules

- Every number in the abstract comes from the paper. If a number changes after the “freeze” from the analysis, the abstract is wrong and so is the poster.
- No citations, no figures, no abbreviations you have not defined.
- Lead Results with your best measured figure, not your composite or model output. A count of people is more persuasive than an index score and always will be.
- Apply the substitution test from §4.2 to the whole abstract. An abstract that could describe any city has not been written correctly.
- Check the word limit and the required headings for each symposium separately. They differ, and an over-length abstract is often rejected without review.

10.3 Worked example

The reference paper's abstract, annotated. Note that Background states four specific policy absences rather than characterizing the environment as "restrictive"; that Methods discloses the projection, the weighting, and the sensitivity design in two sentences; and that Results opens with the largest measured number in the paper and gives four more before stopping.

RESULTS PARAGRAPH, REFERENCE ABSTRACT

"Of 2,613,539 county residents, 2,196,211 (84.03%) live more than one mile from any fixed naloxone access point, 1,506,855 (57.66%) beyond two miles, and 902,611 are stranded beyond both one mile of supply and one-half mile of transit. Exactly one fixed 24-hour access point exists countywide, and 86.25% of residents live more than five miles from it. Of 643 ranked tracts, 170 fall in the highest vulnerability tier, driven more by transit inaccessibility (38.2% of composite) than by modeled mortality (15.2%). Twenty-five optimally selected, non-overlapping storefronts would newly place supply within a half-mile walk of 231,806 currently unserved residents."

11 · DELIVERABLE 8 — THE POSTER

Built Week 12, final Week 15. A SaveSpots template is provided.

You will be given the Dallas poster as an editable PowerPoint file. It is sized 48 × 36 inches landscape, the most commonly accepted large-format size. Confirm your symposium's specification before printing — some cap at 44 × 36 or require portrait.

11.1 Structure

Panel	Contents
Header	Title, authors with superscript affiliations, program line, and a QR code to your preprint DOI
Column 1	Background and significance; the specific data constraint your city presents; objectives; methods including your measured/modeled/published classification
Column 2	Your hero figure, large, with a short caption; the measured results stat band
Column 3	Your second figure; the actionable finding; conclusions; limitations; contact and acknowledgments

11.2 Rules

- **The centerpiece is the measured figure, not the modeled one.** This rule holds across every city regardless of what data you have. A composite index map is visually appealing and requires a paragraph of explanation before it means anything; a map of a directly measured quantity means something at a glance. Dallas leads with measured block-level access, not with the composite.
- Four or five headline numbers maximum in the stat band. A poster with fifteen statistics has none.
- Every figure carries its sources and vintage in the caption, exactly as in the manuscript. A poster is a published artifact.
- The limitations panel is required. It is short — three or four lines — and it is the panel that most often earns respect from a faculty judge.
- Body text at 24pt minimum; section headers at 36pt or larger; title readable from roughly ten feet. Test by viewing at 25% zoom — if you cannot read the body text, neither can anyone standing at your poster.
- The privacy floor in §2.8 applies to posters exactly as it applies to preprints.

11.3 Finding a symposium

Identifying your target venues is your responsibility and it starts in Week 1, not Week 15. Aim for undergraduate research symposia and community-engagement or global-health conferences, which are

accessible and where this work fits well. Your own institution's undergraduate research office is the first place to look; many universities host symposia open to external submissions.

The calendar works in your favor. Most spring symposia (February–April meetings) post their calls between November and January, which is after your numbers are frozen and after your preprint posts. You will be submitting finished work with a citable DOI attached, not promissory results. Bring a target list with deadlines to Week 4.

12 · SEMESTER CALENDAR

Fifteen Sunday meetings. Deliverables are due at the meeting listed. Two dates are gates: Week 4 and Week 10. Nothing passes those without sign-off.

Wk	Date	Meeting focus	Due
—	Tue Jul 28	Kickoff: program overview, standards, tooling workshop	Read Dallas paper
1	Aug 2	City assignment; GitHub and Claude Code set up and verified working; symposium search begins	Repo initialized
2	Aug 9	Data reconnaissance; environment declaration discussed	Data inventory, first pass
3	Aug 16	City profile review; who is missed, and why	Deliverable 1 — City profile
4	Aug 23	SCOPING GATE — analysis plan approved before any pipeline code	Deliverables 2 and 3; symposium target list
5	Aug 30	Pipeline: geography and population layers	Blocks/tracts joined, populations verified
6	Sep 6	Pipeline: supply inventory and its access reality	Supply registry with verification notes
7	Sep 13	Pipeline: primary measured analysis	Question 2 answered, first numbers
8	Sep 20	Composite or prioritization output; sensitivity variants	Sensitivity results reported
9	Sep 27	Placement or targeting output; figure drafts	All four figures drafted
10	Oct 4	ANALYSIS FREEZE — release tagged, Zenodo DOI minted	Deliverable 4 — Repository frozen
11	Oct 11	Manuscript §1–4; abstract drafted from frozen numbers	Manuscript first half; abstract v1
12	Oct 18	Manuscript §5–8; poster build	Manuscript second half; poster v1
13	Oct 25	REVIEW GATE — full manuscript read and returned	Deliverable 5 — Full manuscript
14	Nov 1	Revisions; statements block; submission checklist	Revised manuscript; six statements complete
15	Nov 8	Preprint submitted; poster and abstract finalized	Deliverables 6, 7, 8

12.1 How the review gate works

We'll read every manuscript. The cohort is split in half: Group A submits at Week 12 for a Week 13 return, Group B submits at Week 13 for a Week 14 return. Your group is assigned at the scoping gate the week of the 25th. Reviews are substantive and will require revision; budget for that rather than treating Week 13 as the finish line.

13 · TOOLING

Covered in the kickoff workshop. Set up in Week 1 and verified working before Week 2.

- **GitHub.** One repository per city, yours, public by the freeze. Free.
- **Claude Code.** Strongly recommended and the assumed environment for the workshop and for pipeline support. It is a paid subscription ($20/mo). Let us know if this raises an issue.
- **Zenodo.** Free. Connect it to GitHub once and a DOI is minted automatically on each tagged release.
- **ORCID.** Free, two minutes, required before you submit anything. It is the identifier that makes you legible as a researcher rather than an email address.

13.1 On using AI tools

AI tooling is permitted and expected for pipeline construction, figure generation, drafting, and editing. It is disclosed in the manuscript. There is one hard line:

> **AI IS NOT A SOURCE**
>
> No statistic, policy detail, program outcome, citation, author name, article title, or publication date may originate from an AI-generated summary. Every factual claim must be traceable to a primary source, or to a credible secondary source that cites one — a source you personally opened and read.
>
> Language models fabricate citations that look correct in every surface respect. The reference paper's own readiness audit caught a fabricated citation that had survived multiple drafts: a plausible author, a plausible year, a plausible journal, attached to a real statistic. It was replaced with the verified source and the statistic corrected from 16% to 16.4%.
>
> Check every reference against the actual document before submission. A fabricated citation in a published preprint is permanent, discoverable, and attributable to you by name. It will not be treated as a formatting error.

14 · SOURCE STANDARDS

14.1 The non-negotiable rule

Never fabricate. Do not invent statistics, policy details, program outcomes, quotes, article titles, authors, publication dates, or datasets. If you cannot find a figure from a credible source, say so explicitly and treat it as a data gap. A paper with honest gaps is stronger than one with fabricated precision, and every reader who matters knows the difference.

14.2 Tier 1 — always preferred

- CDC, NIH, SAMHSA, NIDA: national surveillance data and research
- State health departments: vital statistics and overdose surveillance reports
- County health departments: local dashboards and community needs assessments
- U.S. Census Bureau: decennial and ACS
- Peer-reviewed journals: PubMed-indexed, JAMA, AJPH, Harm Reduction Journal, International Journal of Drug Policy

- GAO, KFF, Brookings, Baker Institute and comparable research institutions
- Legal databases and statute text, for any statutory claim

14.3 Tier 2 — acceptable with care

- Local news with a named reporter, named sources, and traceable factual claims
- Established nonprofit harm reduction organizations reporting their own program data
- Official point-in-time homelessness counts

14.4 Prohibited

- Wikipedia, Reddit, Quora, social media of any kind
- Blogs or advocacy posts without a named author and verifiable institutional affiliation
- Any statistic appearing in an article without a cited source for where it came from
- AI-generated summaries used as factual references
- Pay-to-publish data aggregators without methodological documentation

15 · AUTHORSHIP, AFFILIATION, AND WHAT COMES AFTER

15.1 Authorship

The default order on each city paper is: the research assistant first, Ali Ilyas second, Sameer Ilyas third.

That default is earned, not automatic. Authorship requires substantial contribution to the work, participation in drafting or critical revision, approval of the final version, and accountability for it. In practice the second and third positions reflect analytical supervision, pipeline architecture, and substantive review. If a paper receives materially less of that support, the order changes, and if a student's work is genuinely independent, it should.

15.2 Affiliation

Exact wording TBD

15.3 Finding a mentor and pursuing publication

Once your preprint is live you have something most undergraduates emailing a professor do not: a citable DOI, a public repository, and a specific methodological question. That is a real basis for outreach.

What a good cold email contains: two sentences on what you found, one link to the DOI, one specific question about the work, and one clear ask.

Be honest with yourself about the ask. Faculty generally prefer to shape an analysis rather than inherit one, so the strongest framing is forward-looking — the preprint is what you have done, and the question is what the next version should be. A preprint can be revised substantially, including its methodology, before journal submission, so a mentor who wants to reshape the work is not a problem. That is the point.

Target people whose published work is adjacent to yours: spatial epidemiology, harm reduction services research, health geography, or drug policy. Read a paper of theirs before you write. Ali and Sameer can help you with this.

16 · QUALITY STANDARDS

Level	Characteristics
Pretty Good	All sections analytically substantive. Policy analysis consistently distinguishes law from practice. Implementation analysis addresses who is missed. Gaps specific to this city. Recommendations tied to findings. Analysis reproducible from the repository. Limitations honest and specific.
Golden	All of the above, plus: comparison cases precisely calibrated to the city's analytical needs. A genuine methodological contribution with an independence check, a novel measured quantity, a registry others can use. Cultural analysis nuanced and city-specific. Stakeholder leave-behind or deployed map produced at professional quality.

16.1 Common failure modes

Failure	How to correct it
Generic writing	Apply the substitution test to every paragraph. If your city's name can be swapped for another without changing the sentence, rewrite it.
Description without analysis	After every finding, answer: what does this mean specifically for harm reduction access or delivery here? That answer belongs in the paper.
Recommendations not earned by findings	Every recommendation cites the specific gap it addresses and the specific zone or population it targets.
Policy listed without an implementation layer	For every policy: who uses or enforces it, who it fails to reach, and what the distance is between legal status and practical access.
Figure not connected to the text	Reference each figure where it appears, explain what it shows, and connect its findings to the analysis and the recommendations.
Overstated certainty	Distinguish measured from modeled in the sentence itself. Use hedged language where hedging is accurate. Never present an estimate as a count.
Undisclosed weights	Disclose the scheme, justify it in a sentence, run at least two variants, report the flip rate — including when it fails your own threshold.
Coarse data pushed to fine geography	Do not allocate. Exclude the layer and document the exclusion as a limitation.
AI-sourced claims	Every factual claim traceable to a document you opened. Check every reference before submission.

17 · USING THE REFERENCE PAPER

Before submitting any deliverable, read the corresponding section of the Dallas paper and ask whether your version does the same quality of analytical work for your city. Five things specifically repay close study:

Study this	Where	Why it matters for you
How the abstract states a full policy environment in two sentences without losing precision	Abstract, ¶1	Sets the standard for density. Four named absences, not the word "restrictive."
How the methods section handles suppression, proxy use, and the conservative estimate rule	§2.3 and §2.9	The correct way to be honest about what an analysis can and cannot claim. Acknowledging limits strengthens a paper; ignoring them undermines it.
How the implementation analysis moves past program listing into who outreach systematically misses	§5.1–5.5	The difference between "these programs exist" and "here is who they reach, who they do not, and why." That move is required in your paper.
How comparison cases are selected and what is extracted from them	§6	Houston as structurally analogous within the same state regulatory environment; Philadelphia as best practice on comparable

Study this	Where	Why it matters for you
		demographics. Cases chosen to illuminate something specific, producing named lessons rather than descriptions.
How the prioritization connects each tier to an intervention type, and how §8.7 closes by stating the evidentiary limit	§8.1–8.7	Prioritization must produce actionable conclusions. And the closing move — presenting mortality at its honest resolution and naming the gap between that and block-level precision as the paper's central methodological argument — is the strongest single passage in the paper. Find your city's equivalent.